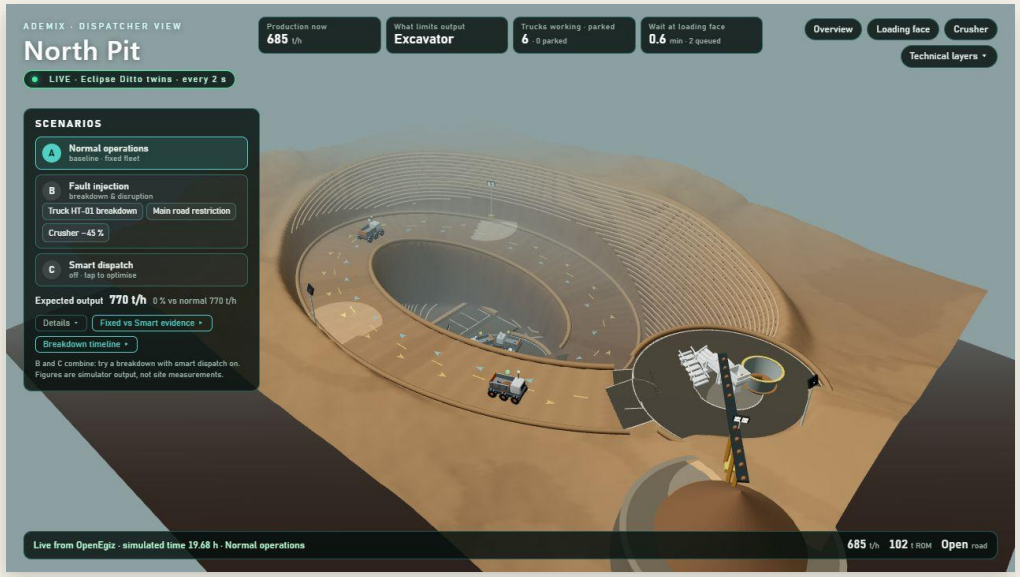
DIGITAL MINE ON OPENEGIZ 2026

North Pit Digital Twin

A live 3D digital twin of an open pit’s loading, haulage and crusher chain on the OpenEgiz platform, built to show the dispatcher what limits output and what to do next.



Live capture of the running stand (Eclipse Ditto)

Built with Eclipse Ditto, MQTT, React Three Fiber and Python

TEAM

Ademix

AUTHORS

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INTRODUCTION

OPERATIONAL CONTEXT AND DECISION PROBLEM

WHO

The mine dispatcher and shift supervisor

The people who decide, minute by minute, where every haul truck goes.

DECISION

Response to disruptions

A truck breaks down, the haul road is restricted, or crusher capacity falls: compare throughput, queues, cycle time and utilization before choosing.

WHY HAULAGE

Haulage links every facility

Face, road and crusher share one flow, so a problem and its consequence are both visible in a live 3D view.

The mine is a **fictional, reduced layout** calibrated from the anonymized North Pit configuration of the MIT-licensed OpenMines project. Synthetic live telemetry is kept separate from sourced parameters.

COMPETITION TASK AND PROJECT RESPONSE

Requirement of “Digital Mine on OpenEgiz”	How the project meets it
Digital twin of a mine facility on OpenEgiz	Open-pit loading, haulage and crusher chain as 11 Eclipse Ditto twins
Simulate equipment operations	Excavator, six 77 t trucks, dump bay and crusher with FIFO queues and cycle timing
Model transportation processes	3.2 km two-lane haul road, speed limits and direction-aware breakdowns
Analyze production data	Live KPIs, matched fixed-versus-smart evidence, recovery history and fleet curves
Predict facility performance	Analytical capacity and bottleneck for every condition, within 0.3 % of the simulation
“What-if” scenarios	Breakdown, road restriction, crusher slowdown and Smart Dispatch, each with a clean reset
Submission	GitHub repository, English, one Docker Compose command, no paid service

INTRODUCTION

PRODUCTION CHAIN: LOADING FACE TO PRIMARY CRUSHER

LOADING FACE

770 t/h

One excavator; four 20.32 t passes load a 77 t truck in 6 min

HAUL ROAD

3.2 km

Two separated lanes, 25 km/h, six 77 t haul trucks

ROM BIN

600 t

One dump bay; a truck tips only when a full load fits

CRUSHER

900 t/h

Primary crusher, planning capacity

Sourced: 77 t payload, 20.32 t shovel pass (OpenMines) **Derived**: 6 min loading

Team assumption: 3.2 km route, 600 t bin, 900 t/h crusher

PURPOSE

RESEARCH AIM AND OBJECTIVES

> Give the mine dispatcher a live digital twin that shows what limits ore flow, and which fleet decision restores it, with evidence that can be checked.

RESEARCH OBJECTIVES

01

Identify the binding constraint, loader, road or crusher, under each operating condition.

02

Determine the smallest fleet that sustains maximum output in each condition.

03

Quantify the production loss and the recovery after an equipment breakdown.

TASKS

RESEARCH TASKS AND IMPLEMENTATION STATUS

#	Task	Done, and where to see it
1	Represent the chain as 11 connected Eclipse Ditto twins	Mine model view
2	Simulate it at 20× speed: FIFO queues, cycle times, mass balance	98 simulator, API and evidence tests
3	Publish all 11 twins through MQTT to Ditto every 5 s; the frontend polls every 2 s	LIVE badge, smoke check
4	Visualize it as a 3D mine driven by twin state	Command center
5	Run three disruptions with a clean reset	Scenario controls B, reset A
6	Turn analysis into a decision the dispatcher can apply	Smart Dispatch, Decision
7	Prove it with tests, matched evidence and stated limits	Model credibility panel

METHOD

SYSTEM ARCHITECTURE ON OPENEGIZ

1. Simulator

Deterministic Python engine: FIFO queues, exclusive loader and dump bay, mass balance

→

2. MQTT

Ditto Protocol updates: a complete 11-twin snapshot every 5 seconds

→

3. Eclipse Ditto

11 digital twins, the single source of state

→

4. Ademix API

Frontend polls every 2 s; normal Ditto cache TTL: 3 s. Scenario and dispatch controls.

→

5. 3D view

React Three Fiber command center labels LIVE, DEMO or REPLAY

11

twins in Ditto

143

required fields with units, checked by smoke test

5 s

simulator publishes a complete snapshot

2 s

frontend polls the Ademix API

20×

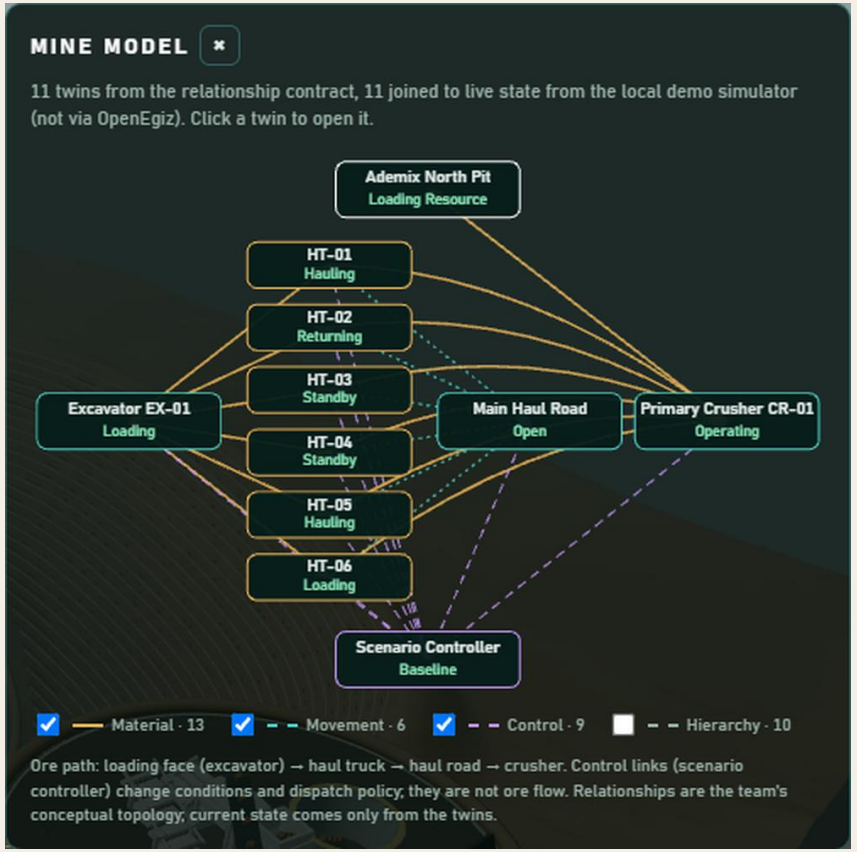
simulated seconds per real second

METHOD

DIGITAL TWIN MODEL AND RELATIONSHIPS

Twins	Kind
6	haul trucks, 77 t
1	excavator and loading face
1	haul road, two lanes
1	crusher with dump bay and ROM bin
1	scenario controller
1	mine aggregate, the parent

Links: **13 material**, **6 movement**, **9 control**, **10 hierarchy**. Control links never pretend to be ore flow; clicking a node opens that live twin.



Mine model view (local demo mode, same engine)

METHOD

FUNCTIONAL MODULES OF THE TWIN

Fleet supervision
Each truck’s state, payload, speed, trips and cycle time

Production monitor
Throughput, live constraint, queues and ROM bin level

What-if engine
Three disruptions started and reset from the stand

Smart Dispatch
Sizes the fleet to the bottleneck and parks surplus trucks

Decision workspace
Recommendation, evidence and a confirmed Apply control

Show effect
Focuses the affected assets and explains the causal chain

Mine model explorer
The 11 twins and 38 typed links, each node clickable

Fuel and CO_2 estimate
Shift diesel and tank-to-wheel CO_2, labelled estimates

Guided demonstration
Six steps on real controls, each confirmed by the twins

Also available as technical layers: equipment specifications, IoT sensors, topology, data flow, weather and safety.

RESULTS

WHAT-IF SCENARIO ANALYSIS

Condition	Output t/h	Cycle	Face wait	Crusher wait	What the dispatcher should do
Normal	770.0	36.0	13.6	0.0	Loader limits output: park surplus trucks
HT-01 breakdown	770.0	30.1	7.6	0.0	Tow the blocked lane; five trucks still feed the loader
Road at 10.72 km/h	645.3	42.8	0.0	0.0	Restore road speed; keep all six trucks cycling
Crusher −45 %	495.0	51.6	3.4	25.4	Restore the crusher; park trucks to limit dump queueing

Hours 2–9 after a common four-hour warm-up, from compare_scenarios.py. Cycle and waits in minutes. Every scenario starts and resets from the stand without code changes.

RESULTS

PREDICTED CAPACITY MATCHES THE SIMULATION

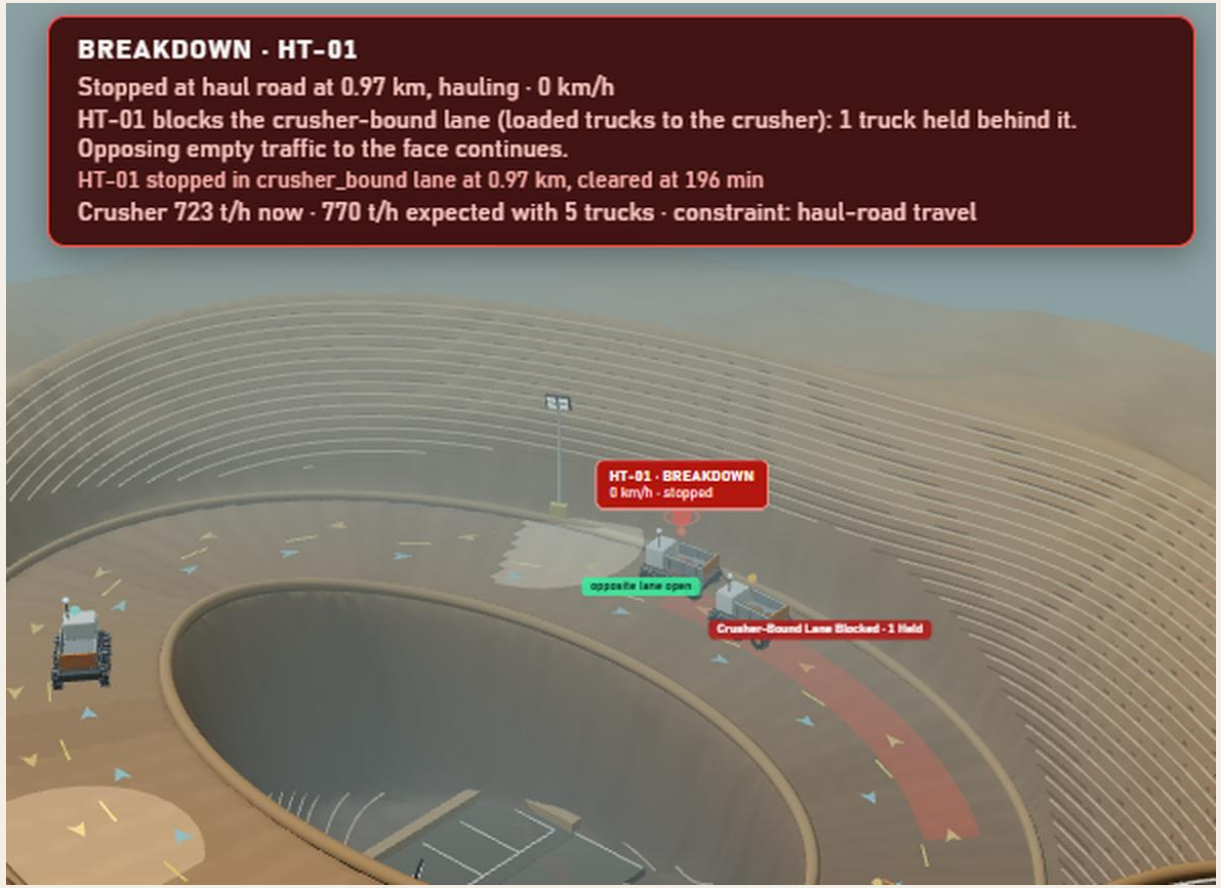
```
capacity = min(fleet supply, loading 770, dump 4,620, crusher 900 or 495) t/h
```

Condition	Predicted constraint	Predicted t/h	Simulated t/h	Gap
Normal	Loading	770	770.0	0.0 %
HT-01 breakdown	Loading	770	770.0	0.0 %
Road at 10.72 km/h	Haul-road travel	647.4	645.3	−0.3 %
Crusher −45 %	Crusher	495	495.0	0.0 %

The predicted constraint equals the live, measured constraint in all four conditions. Simulated values: hours 2–9 after a four-hour warm-up.

RESULTS

TRUCK BREAKDOWN: LANE-SPECIFIC CAUSE AND EFFECT



1. HT-01 stops at 0 km/h on the haul road.
2. Only its own lane is blocked; opposing traffic keeps moving.
3. Towed clear after 20 simulated minutes.
4. First hour: 770 → 691 t/h, −10.3 %.

Capture: local demo mode, same engine

RESULTS

SMART DISPATCH: EQUAL OUTPUT, FEWER TRUCK-HOURS

6 → 4

trucks cycling; two park as ready reserves

770

t/h with either policy

–33 %

truck-hours per 1,000 t (7.79 → 5.19)

1.6

min wait at the face, down from 13.6

Truck-hours per 1,000 t, Smart versus fixed, by condition:

- Normal **–33 %**
- Breakdown **–20 %**
- Road restricted **0 %**
- Crusher slow **–50 %**

Matched 8-hour windows from one identical start. Estimated diesel per 1,000 t falls 26 % in Normal. Model estimates, not measured savings.

RESULTS

SUMMARY OF VERIFIED RESULTS

143

required fields with units across 11 Eclipse Ditto twins, all returned live

8 of 8

fixed and Smart runs across all four conditions pass live, then reset cleanly

0 t

ore mass-balance error in every scenario: no tonne created or lost

−33 %

truck-hours per 1,000 t at the same 770 t/h; −50 % under crusher slowdown

−79 t

total cost of a breakdown over 3 h, then output returns to 770 t/h

146

tests passed: 98 simulator, API and evidence, plus 48 data tests

Clean Ubuntu 24.04 AMD64 verification

3 min 33 s

full-stack startup

11 twins

Eclipse Ditto integration verified

Passed

full scenario matrix and Smart Dispatch

Passed

Docker restart and post-restart smoke

Savings and losses are model results, not mine measurements.

OUTCOME

SCIENTIFIC AND TECHNICAL NOVELTY

1. Decision-oriented twin

The recommendation comes from versioned evidence, is applied through the real control, and shows “in force” only when the twins confirm it.

2. Bottleneck-aware fleet sizing

Trucks needed = capacity × free cycle ÷ payload, rounded up; surplus trucks park and return when a disruption needs them.

3. Direction-aware breakdown model

A failed truck blocks only its own lane; opposing traffic keeps moving, and the 3D view marks the exact lane from twin fields.

4. Matched-start comparative evidence

Every comparison starts from one identical simulated state with identical windows, plus fleet curves and one-at-a-time sensitivity.

5. Explicit credibility layer

Every number is tagged live or saved model evidence; mass balance passes only within 0.000001 t; the limits are stated inside the app.

6. Typed relationship graph

38 links typed as material, movement, control or hierarchy; each node opens its live twin in the 3D mine.

OUTCOME

COMPLIANCE CONTEXT: KAZAKHSTAN SAFETY RULES

Regulatory requirement

- The mine’s project documentation sets safe bench heights, berm and working-platform dimensions and slope angles for the specific site.
- The working-face passport states the permitted dimensions at each face.

Implementation in the twin

Illustrative value in the 3D pit	Value
Bench height	10 m
Bench face angle	70°
Working platform	20 m
Overall slope	45°

The app labels them **“Illustrative assumptions for a conceptual project, not approved Kazakhstan design values.”** On a real site they are replaced by the approved project documentation and face passports, and checked by a qualified mining engineer.

Source: Kazakhstan industrial safety rules for open-pit mining

CONCLUSION

CONTRIBUTIONS AND NOVELTY OF THE PROJECT

Contributions

- 11 live Eclipse Ditto twins of a full face-to-crusher chain
- A deterministic simulation with FIFO queues and 0 t mass-balance error
- A 3D command center drawn only from returned twin state
- Three disruptions, Smart Dispatch and a Decision workspace

Novelty

- Decisions applied through the real control, confirmed by the twins
- Fleet sized to the bottleneck: same 770 t/h, 33 % fewer truck-hours
- Direction-aware breakdowns and matched-start evidence
- A truth layer: every number tagged live or saved evidence

Scope: a deterministic conceptual model on synthetic telemetry, verified against its formulas and not validated against an operating mine.